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DATABASE MANAGEMENT SYSTEMS

1-Introduction and Basic Concepts

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What is data?

- Data is any value (number, letter, character, image, sound, etc.) obtained from measurements, observations, etc. made in the environment and not subjected to any processing.
- When data is organized/processed in a meaningful way, it produces meaningful information (information, knowledge).
- In everyday life, there is a lot of data generated within various systems. For example, blood values taken under constant observation, temperature values measured at different times of the day, etc.

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What is a database?

- It is a collection of data related to a specific subject and stored in an organized manner.
- A database is **an organized collection** of data **(Wikipedia)**
- A database is **a structured collection** of data designed for **efficient storage**, **retrieval**, and **manipulation** (web)
- A database is an organized collection of structured data, typically stored **electronically in a computer system** (web)
- A database is a digital repository for **storing, managing**, and **securing organized collections** of data. (IBM)

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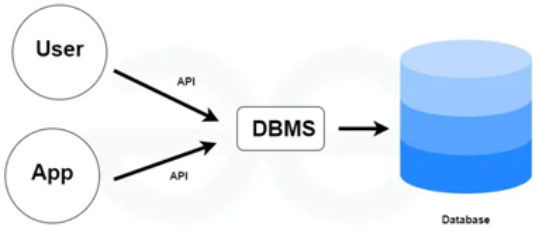
Why a Database?

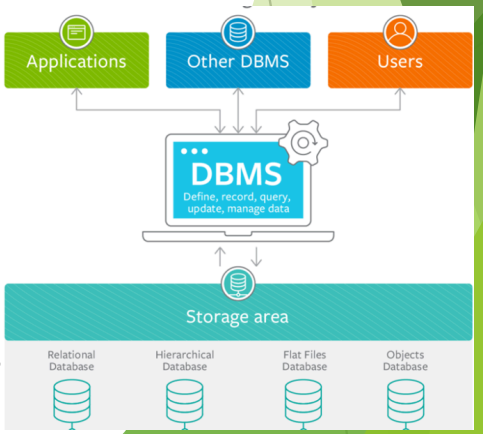
- In the past, information was stored in traditional filing systems. However, data groups to be stored had to be kept separately in different files (safe, bank, sales stock).
- In databases, large amounts of data with different characteristics can be stored in a single structure and in an organized manner, allowing for easy access when needed.
- There are two types of file access methods in this type of filing process.
  - Sequential
    - When searching for information, the file is read from the beginning until the end. For example, music cassettes. This is advantageous if all records are to be processed sequentially, but disadvantageous if specific information needs to be accessed directly.
  - Direct Access
    - A unique and usually sequential index number is generated for each stored line and kept in a separate file. The index value of each line and the disk address corresponding to this index value are found here. The address is learned from this file according to the searched index value and the location is determined.

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Database Management System (DBMS)/ Database Management System (DBMS)

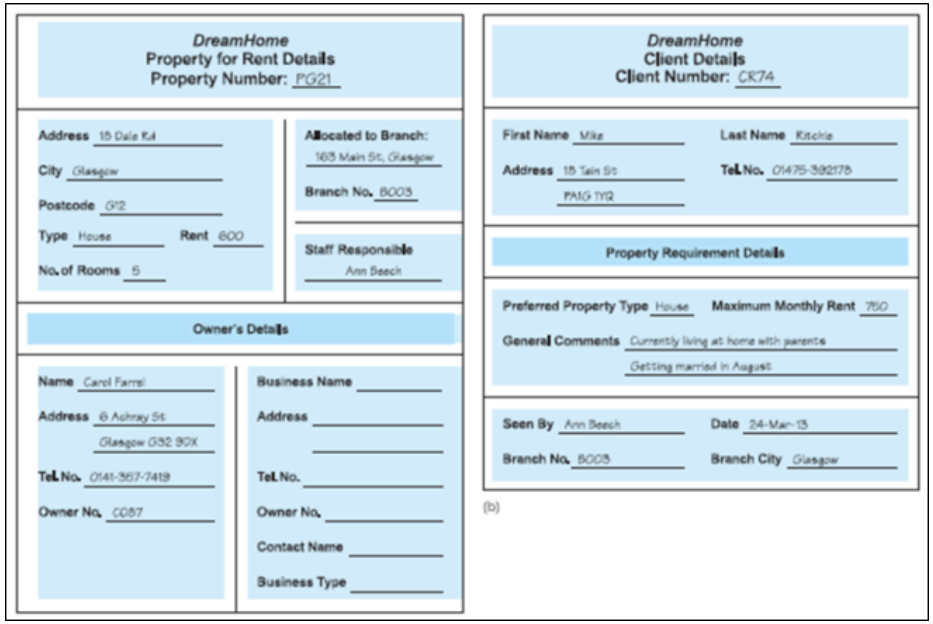
- These are interfaces/systems that enable users to easily perform management operations on the database (such as creating databases, creating tables, performing record operations, queries, security settings, backup, and maintenance).
- There are many DBMSs on the market with different names. Some of these are MS Access, Oracle, MySQL, Sybase, Paradox, MS SQL Server...





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https://en.wikipedia.org/wiki/Comparison_of_relational_database_management_systems



DreamHome
Property for Rent Details
Property Number: PG21

Address 15 Dale Rd
City Glasgow
Postcode G12
Type House Rent 600
No. of Rooms 5

Allocated to Branch:
163 Main St, Glasgow
Branch No. B003

Staff Responsible
Ann Beech

Owner's Details

Name Carol Farrel
Address 6 Achray St
Glasgow G32 9DX
Tel. No. 0141-357-7419
Owner No. CO87

Business Name
Address
Tel. No.
Owner No.
Contact Name
Business Type

DreamHome
Client Details
Client Number: CR74

First Name Mike Last Name Ritchie
Address 18 Tain St
PA1G 1YQ
Tel. No. 01475-392178

Property Requirement Details

Preferred Property Type House Maximum Monthly Rent 750
General Comments Currently living at home with parents
Getting married in August

Seen By Ann Beech Date 24-Mar-13
Branch No. B003 Branch City Glasgow

(b)

TABLE 1.1 Limitations of file-based systems.

Separation and isolation of data
Duplication of data
Data dependence
Incompatible file formats
Fixed queries/proliferation of application programs

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead Rd	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Client

clientNo	fName	lName	address	telNo	prefType	maxRent
CR76	John	Kay	56 High St, London SW1 4EH	0207-774-5632	Flat	425
CR56	Aline	Stewart	64 Fern Dr, Glasgow G42 0BL	0141-848-1825	Flat	350
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178	House	750
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720	Flat	600

Figure 1.2 The PropertyForRent, PrivateOwner, and Client files used by Sales.

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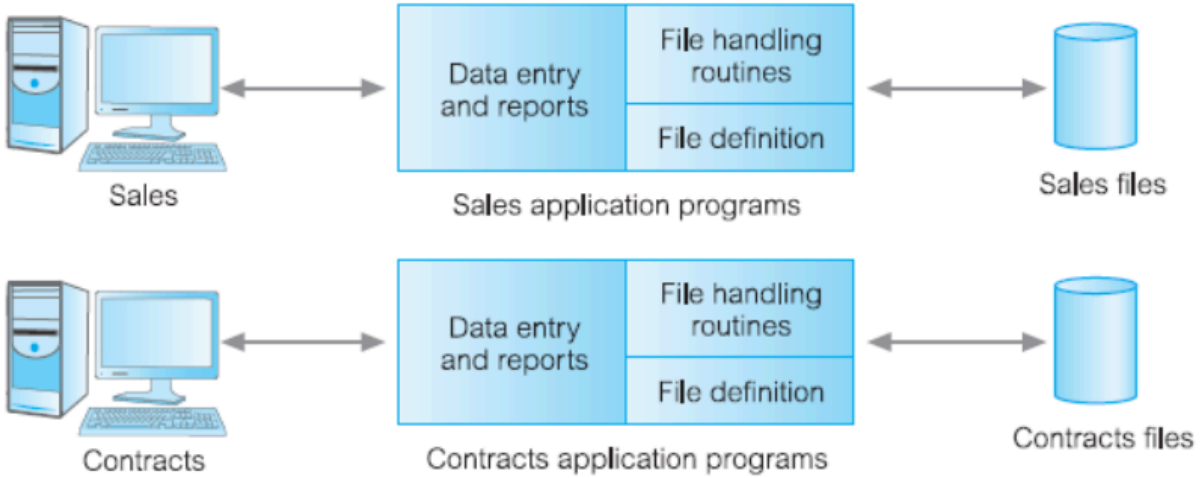


Figure 1.5
File-based processing.

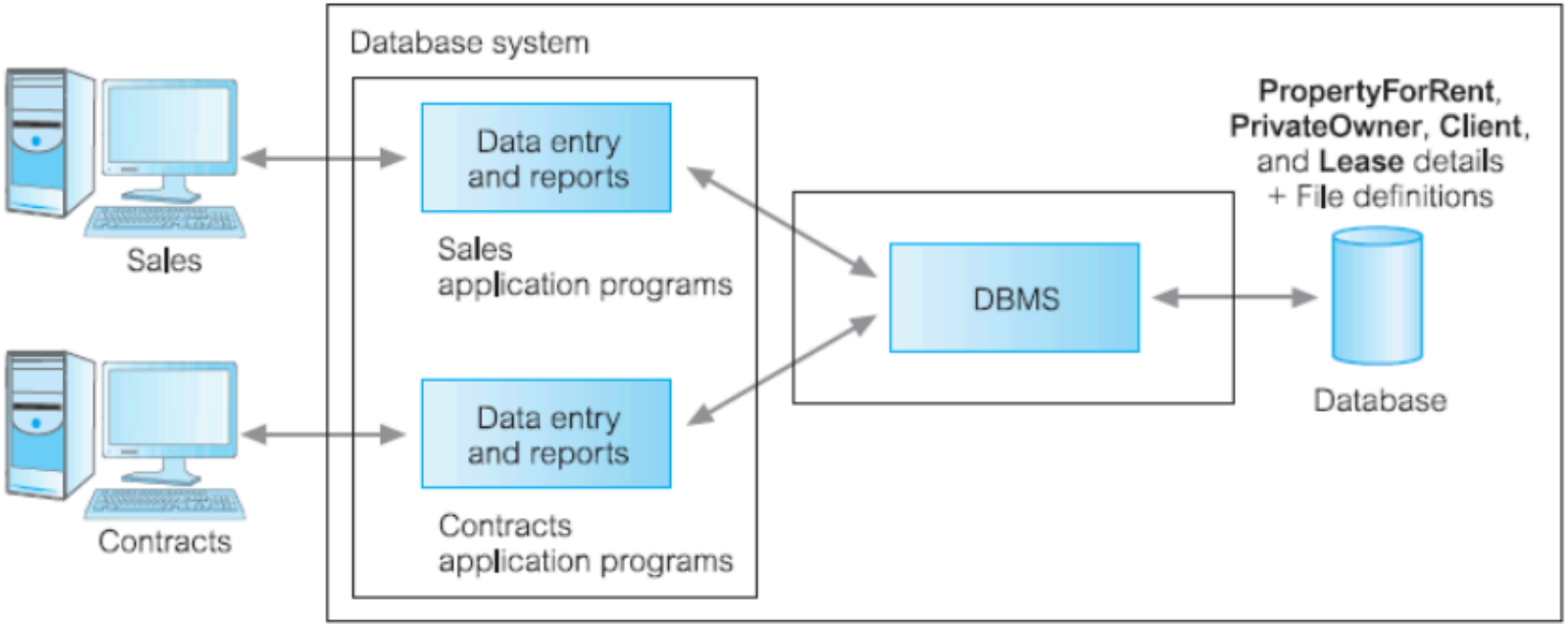
Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

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PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

Figure 1.7 Database processing.

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TABLE 1.2 Advantages of DBMSs.

Control of data redundancy	Economy of scale
Data consistency	Balance of conflicting requirements
More information from the same amount of data	Improved data accessibility and responsiveness
Sharing of data	Increased productivity
Improved data integrity	Improved maintenance through data independence
Improved security	Increased concurrency
Enforcement of standards	Improved backup and recovery services

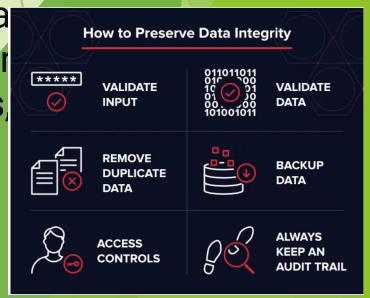
TABLE 1.3 Disadvantages of DBMSs.

Complexity
Size
Cost of DBMSs
Additional hardware costs
Cost of conversion
Performance
Greater impact of a failure

Data Consistency: Data consistency refers to the accuracy and uniformity of data stored across multiple data sources. In a consistent database, once a transaction has been committed, all subsequent accesses to that data will reflect the updated value.[1]

Suppose you have a customer's address stored in two different places in your system – once in a table for billing information, and once in a table for shipping information. If the customer updates their address, and this change is ref ected in the billing information but not the shipping information, you have a consistency problem.

Data Integrity: Data integrity is the preservation and assurance of the accuracy and consistency of data throughout its entire lifecycle. It involves ensuring the validity and accuracy of data and the processes for protecting it. It is a critical element in the design, implementation, and use of any system that stores, receives, or processes data. [2]



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Data Concurrency: Enabling multiple users to access data simultaneously.

1 - https://www.montecarlodata.com/blog-data-consistency/
2 - https://medium.com/peopleofpapara/veri-b%C3%BCt%C3%BCnl%C3%BC%C4%9F%C3%BC-data-integrity-34034c0878d9

TIMEFRAME	DEVELOPMENT	COMMENTS
1960s (onwards)	File-based systems	Precursor to the database system. Decentralized approach: each department stored and controlled its own data.
Mid-1960s	Hierarchical and network data models	Represents first-generation DBMSs. Main hierarchical system is IMS from IBM and the main network system is IDMS/R from Computer Associates. Lacked data independence and required complex programs to be developed to process the data.
1970	Relational model proposed	Publication of E. F. Codd's seminal paper "A relational model of data for large shared data banks," which addresses the weaknesses of first-generation systems.
1970s	Prototype RDBMSs developed	During this period, two main prototypes emerged: the Ingres project at the University of California at Berkeley (started in 1970) and the System R project at IBM's San José Research Laboratory in California (started in 1974), which led to the development of SQL.
1976	ER model proposed	Publication of Chen's paper "The Entity-Relationship model—Toward a unified view of data." ER modeling becomes a significant component in methodologies for database design.
1979	Commercial RDBMSs appear	Commercial RDBMSs like Oracle, Ingres, and DB2 appear. These represent the second generation of DBMSs.
1987	ISO SQL standard	SQL is standardized by the ISO (International Standards Organization). There are subsequent releases of the standard in 1989, 1992 (SQL2), 1999 (SQL:1999), 2003 (SQL:2003), 2008 (SQL:2008), and 2011 (SQL:2011).
1990s	OODBMS and ORDBMSs appear	This period initially sees the emergence of OODBMSs and later ORDBMSs (Oracle 8, with object features released in 1997).
1990s	Data warehousing systems appear	This period also see releases from the major DBMS vendors of data warehousing systems and thereafter data mining products.
Mid-1990s	Web–database integration	The first Internet database applications appear. DBMS vendors and third-party vendors recognize the significance of the Internet and support web–database integration.
1998	XML	XML 1.0 ratified by the W3C. XML becomes integrated with DBMS products and native XML databases are developed.

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Figure 1.10 Historical development of database systems.

List of Relational DBMSs

https://en.wikipedia.org/wiki/List_of_relational_database_management_systems

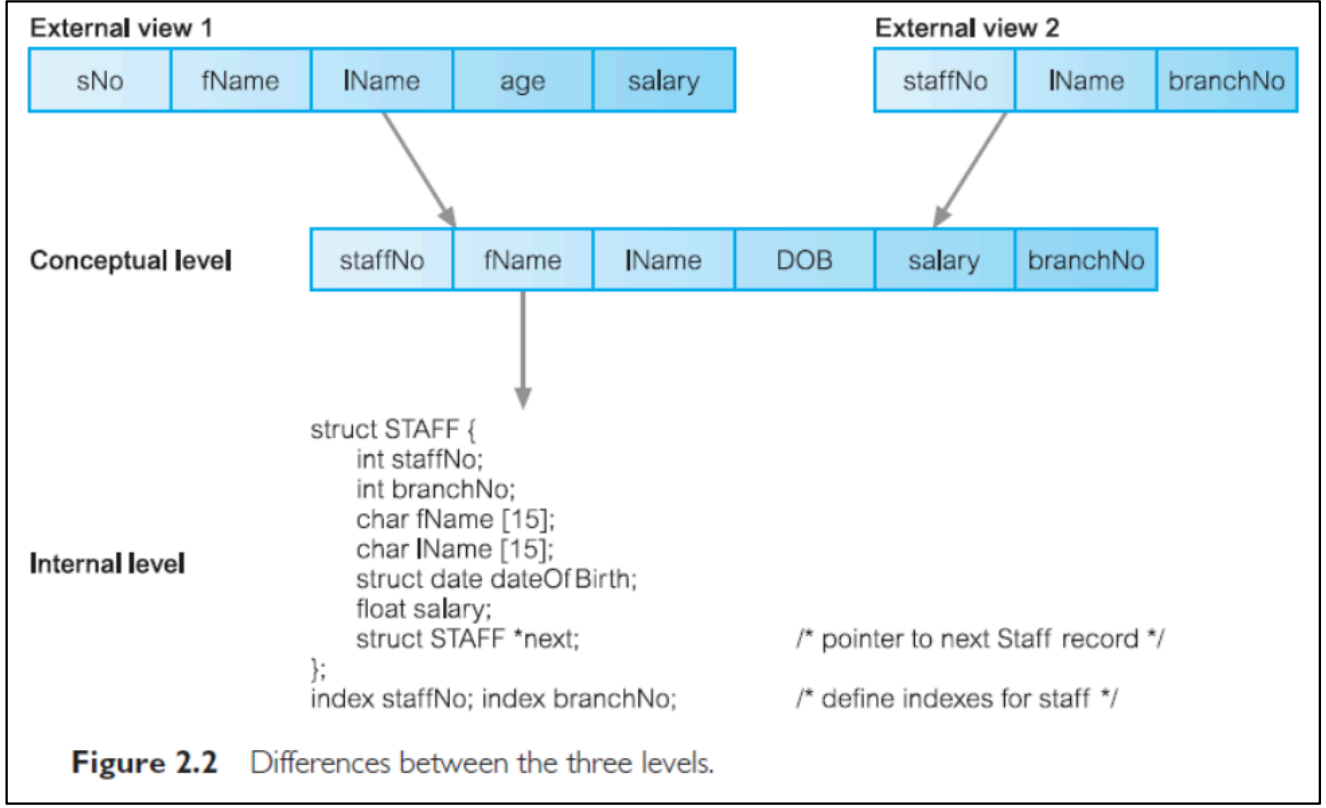


Figure 2.2 Differences between the three levels.

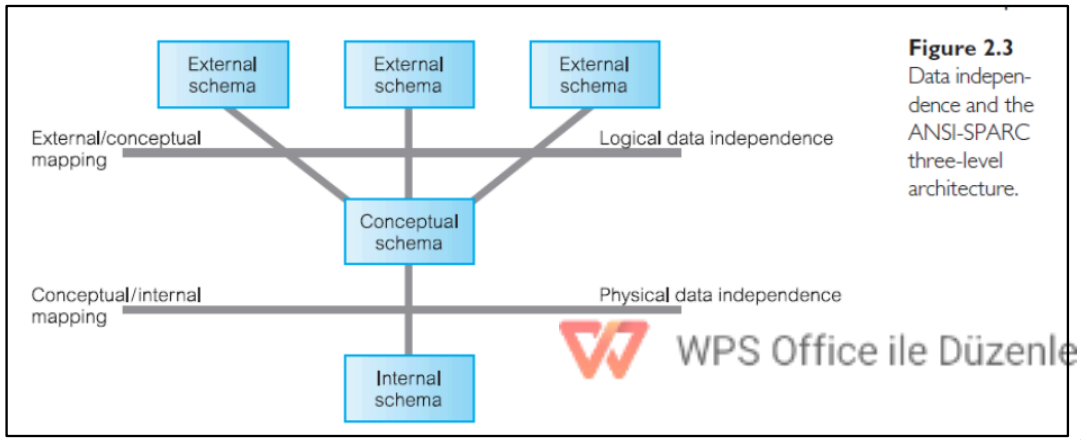
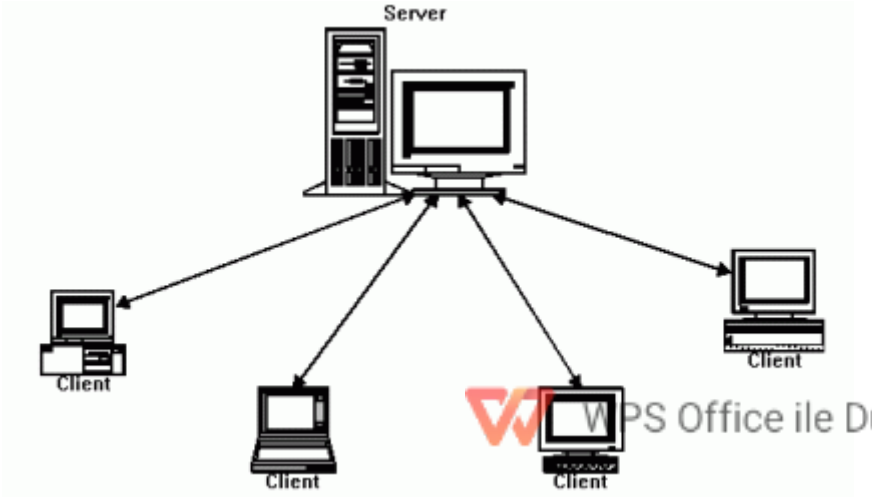


Figure 2.3 Data independence and the ANSI-SPARC three-level architecture.

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Client/Server Architecture

- Client/Server architecture is a centralized structure that allows information or various services to be stored on a single computer (server) in a network environment and enables those who need this information/service to access and use it.
- Management, backup, maintenance, and other processes related to the information/services are easier to perform from a single point.



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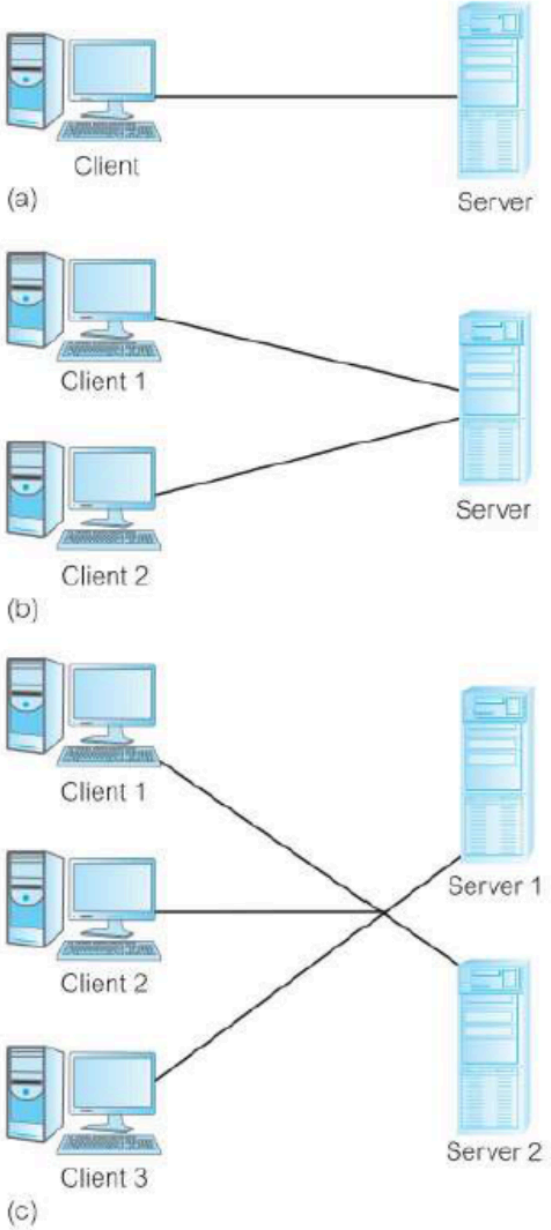


TABLE 3.1 Summary of client–server functions.

CLIENT	SERVER
Manages the user interface	Accepts and processes database requests from clients
Accepts and checks syntax of user input	Checks authorization
Processes application logic	Ensures integrity constraints not violated
Generates database requests and transmits to server	Performs query/update processing and transmits response to client
Passes response back to user	Maintains system catalog Provides concurrent database access Provides recovery control

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Benefits of DBMS

- Data redundancy is prevented.
  - Prevents the same information entered into one table from being re-entered into the same or different tables. Data is stored only once.
- Data consistency is ensured.
  - Changes to data must be reflected everywhere it is stored.
- Data sharing is ensured.
  - Data that is being stored can be made available for use by multiple clients at the same time. When multiple users want to make changes to the same data at the same time, the DBMS prioritizes based on criteria such as privilege hierarchy or connection priority.

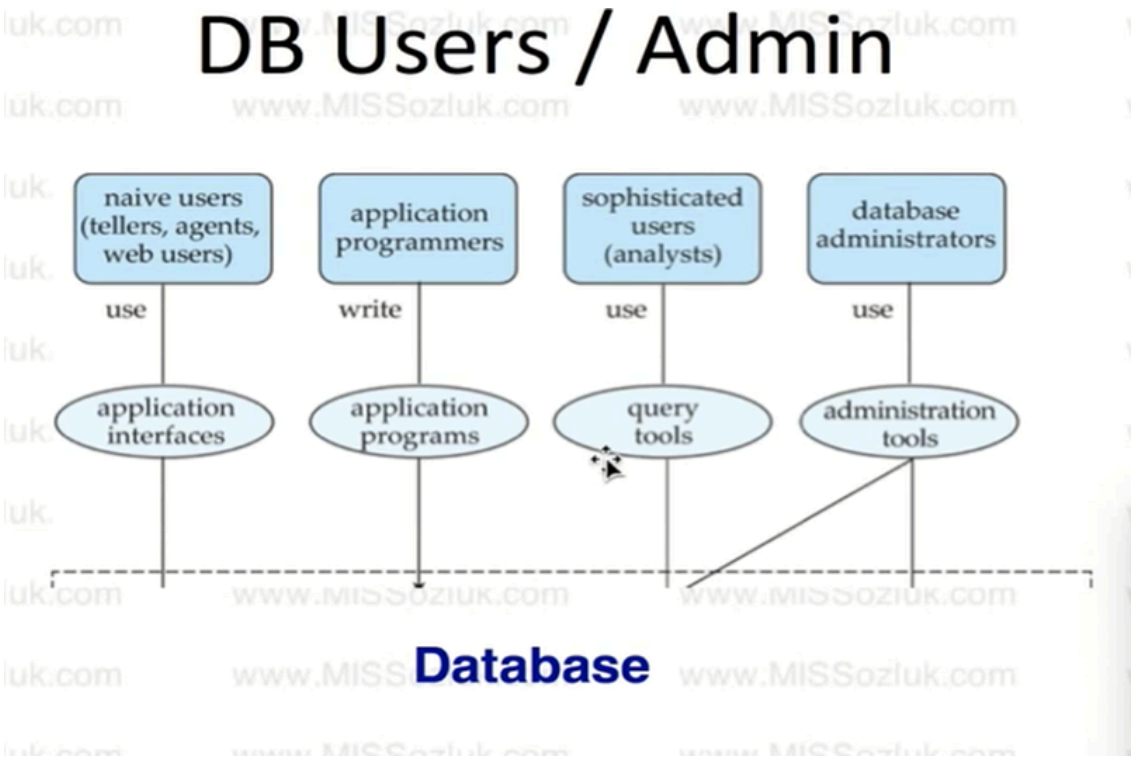
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Benefits of DBMS

- Data integrity is ensured.
  - Prepared or calculated data that is attempted to be written internally must be compatible with the internal table structure (it must be suitable for constraint types and definitions).
- Data security is ensured.
  - Preventing unauthorized users from connecting to the database, ensuring that connected users can only view data within their permissions, and allowing them to perform operations on that data.
- Data independence is ensured.
  - The physical structure of the database is hidden from users. It allows operations to be performed on data by connecting with various programming languages.

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Database Users



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Data Models

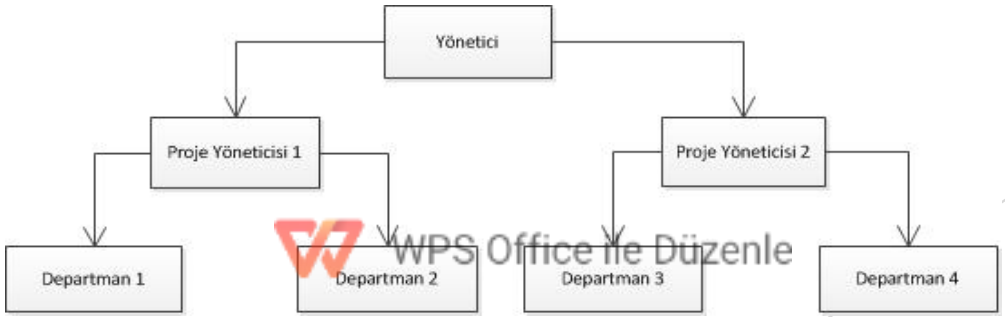
- Although the primary function of databases is to store requested information, they contain structural differences. Structural details such as data storage, processing, and establishing relationships between data are referred to as **the Data Model**.
- Every DBMS uses a data model to organize data logically.
- Examples of data models developed from the past to the present
  - Hierarchical Data Model
  - Network Data Model
  - Relational Data Model
  - Object-Oriented Data Model

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Data Models

Hierarchical Data Model

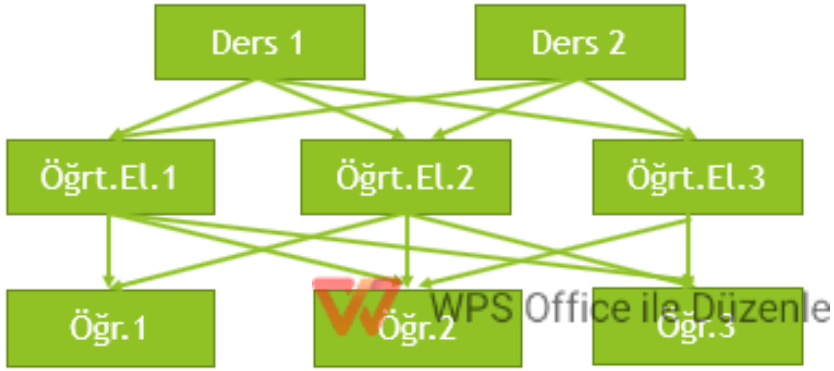
- Used in the 1960s and 1970s. In this model, records are stored in a tree structure.
- Branching and diversity increase thanks to the sub-branches of each branch. A sub-branch must be connected to the parent branch from only one point. It is based on a parent-child relationship. (A parent can have multiple children, but a child has only one parent). In this model, a node can be connected to n nodes at the lower level and 1 node at the upper level.
- When establishing relationships between entities, each entity must be connected to only one other entity. If the same entity is to be used in multiple relationships, it must be recreated, leading to unnecessary data duplication.
- When searching for any information, the search starts from the top.
- If any branch is deleted, all records connected to that branch are also deleted.



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Data Models

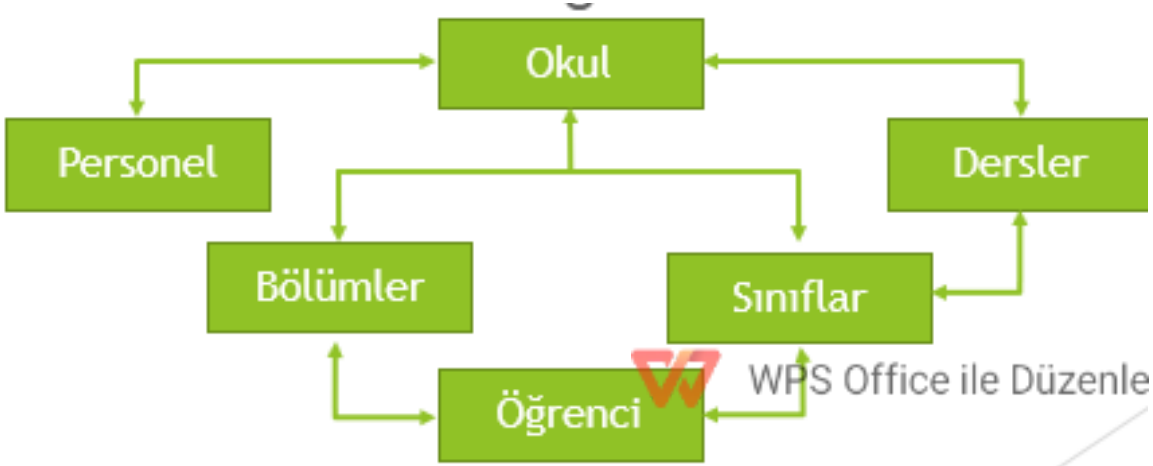
- Network Data Model
  - In a hierarchical model, a node can be connected to n nodes at a lower level and 1 node at a higher level; in a network data model, there is no such limitation. An element can be connected to any other element. Each record can have multiple parent and child records.
  - Since connections between nodes are established using pointers, the increased use of pointers as the structure grows can lead to complex situations in applications and make management difficult.
  - The design and structure of the model are not user-friendly, and tracking becomes difficult as the structure grows



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Data Models

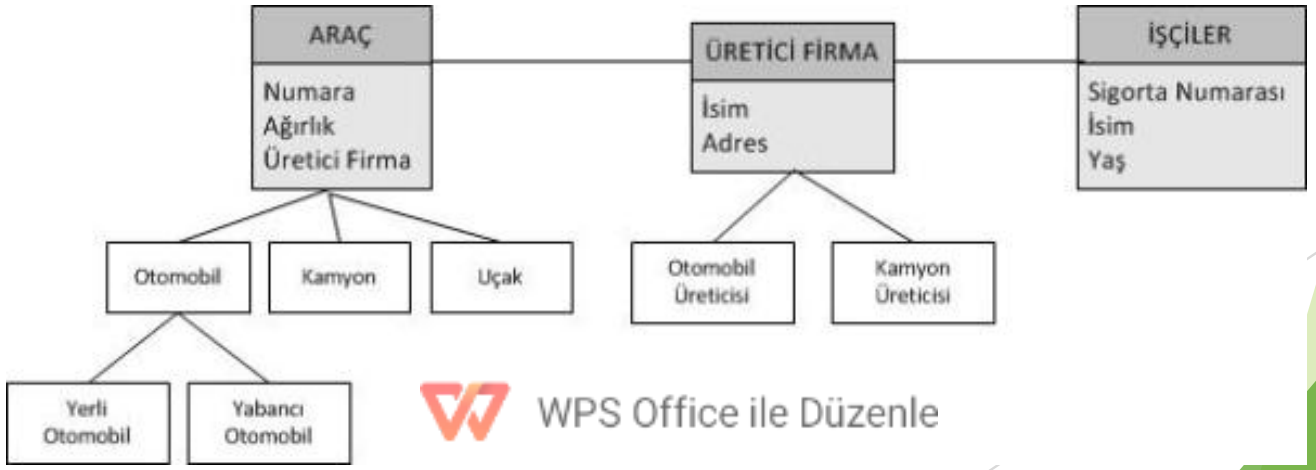
- Relational Data Model
  - A data model that emerged around the 1980s. (E.F. Codd)
  - Although it is not the most recent data model, it is still widely used in many scenarios today.
  - Data with common characteristics are stored in tables. Tables consist of rows and columns. Data and relationships are defined on tables, and all information is visible.
  - When establishing relationships, primary keys (Primary Key-PK) and foreign keys (Foreign Key-FK) are used.



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Data Models

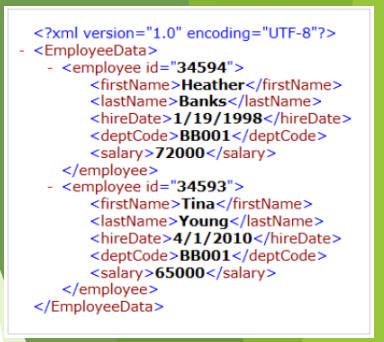
- Object-Oriented Data Model
  - It emerged in the 1990s. The concept of relationship in the relational model corresponds to the concept of class here. Data is modeled and created as objects. The concepts of class and inheritance in Object-Oriented Programming (OOP) also apply here.
  - It is an approach that offers high performance when processing complex data.



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New Data Models

- **Object/Relational Model:** It emerged as a result of combining the relational model with the object-oriented model.
- **XML:** Primarily a data definition standard used for cross-platform data exchange. It is also used to define unstructured data.
- **JSON:** Primarily, it is a data definition standard used for cross-platform data exchange. It is also used to define unstructured data.
- **NoSQL:** ???
- **NewSQL:** ???





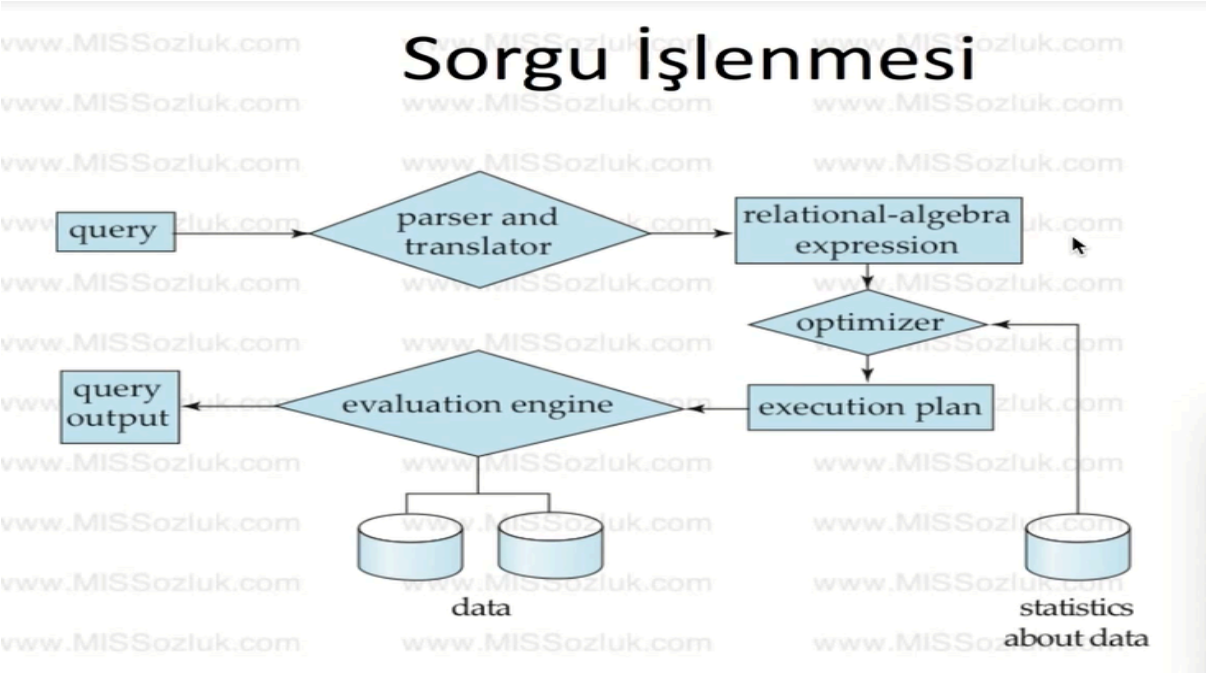
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Basic Database Concepts (Relational Model Database Concepts)

Field	Relation
Data Type	Null
Record / Row	Result Set
Column	Table
Constraint Primary Key Foreign Key Unique Key Default Check	Index Clustered (Physical Order) Non-Clustered (Non-physically ordered)

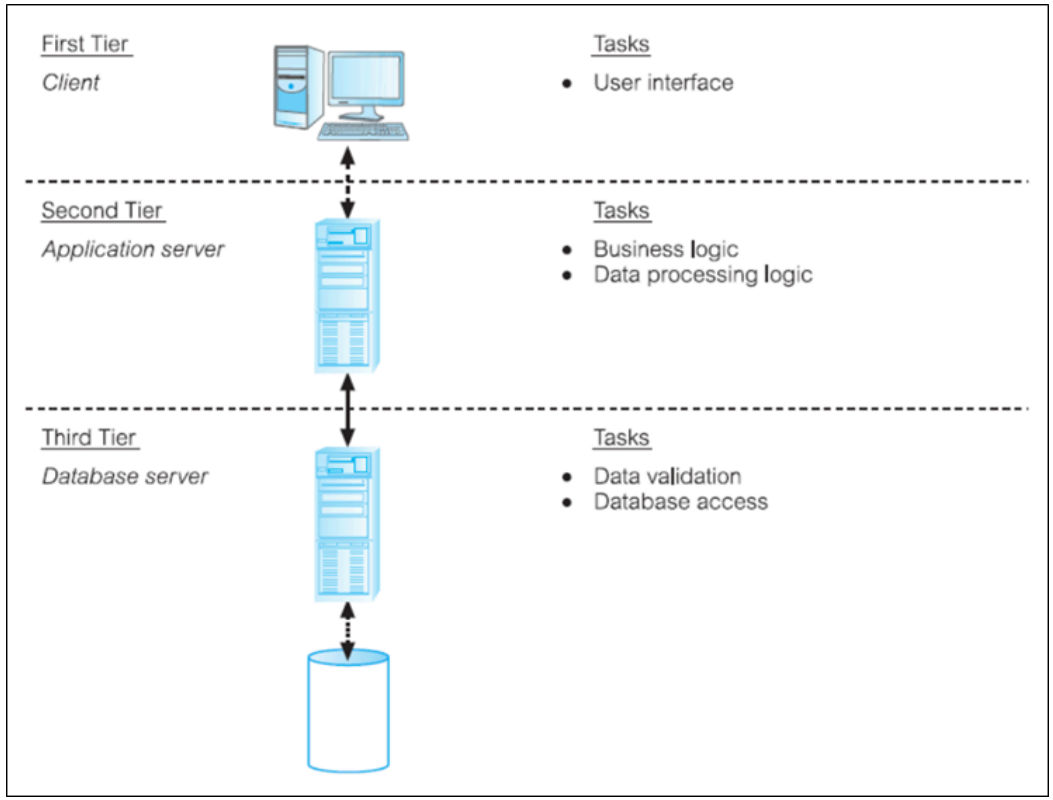
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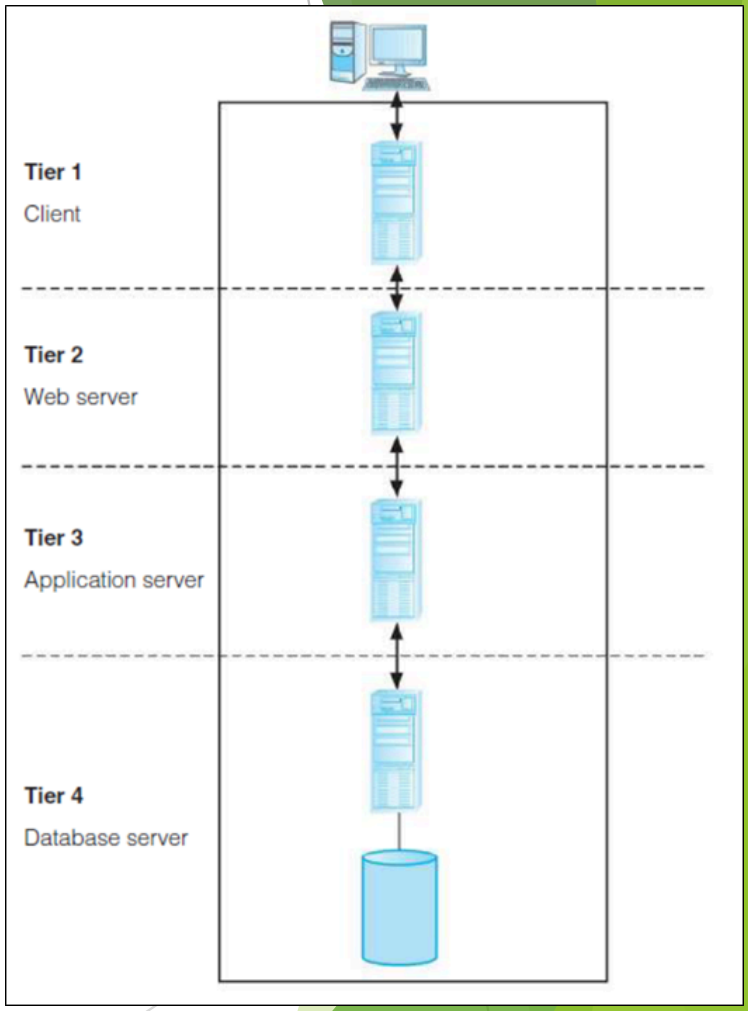
Sorgu İşlenmesi



- **Parser:** Examines the written SQL statement. Checks whether the words it contains, such a s "Select, From, *, Where, data type, defined command or function..." etc., are valid words. If there is an incorrect sequence that violates the required order, it detects this and issues a warning.
- **Translator:** Translates the written SQL statement into relational algebra equations.
- **Optimizer:** Examines the written SQL statement, corrects any errors made within it, creates the most efficient form for this query to run, and creates an "execution plan" for it. (For example, if we consider a query that will retrieve data from two tables, should we retrieve these data separately or run it by combining them into a single table, etc.)
- **Evaluation Engine:** The execution plan created is executed by the database engine at this stage.

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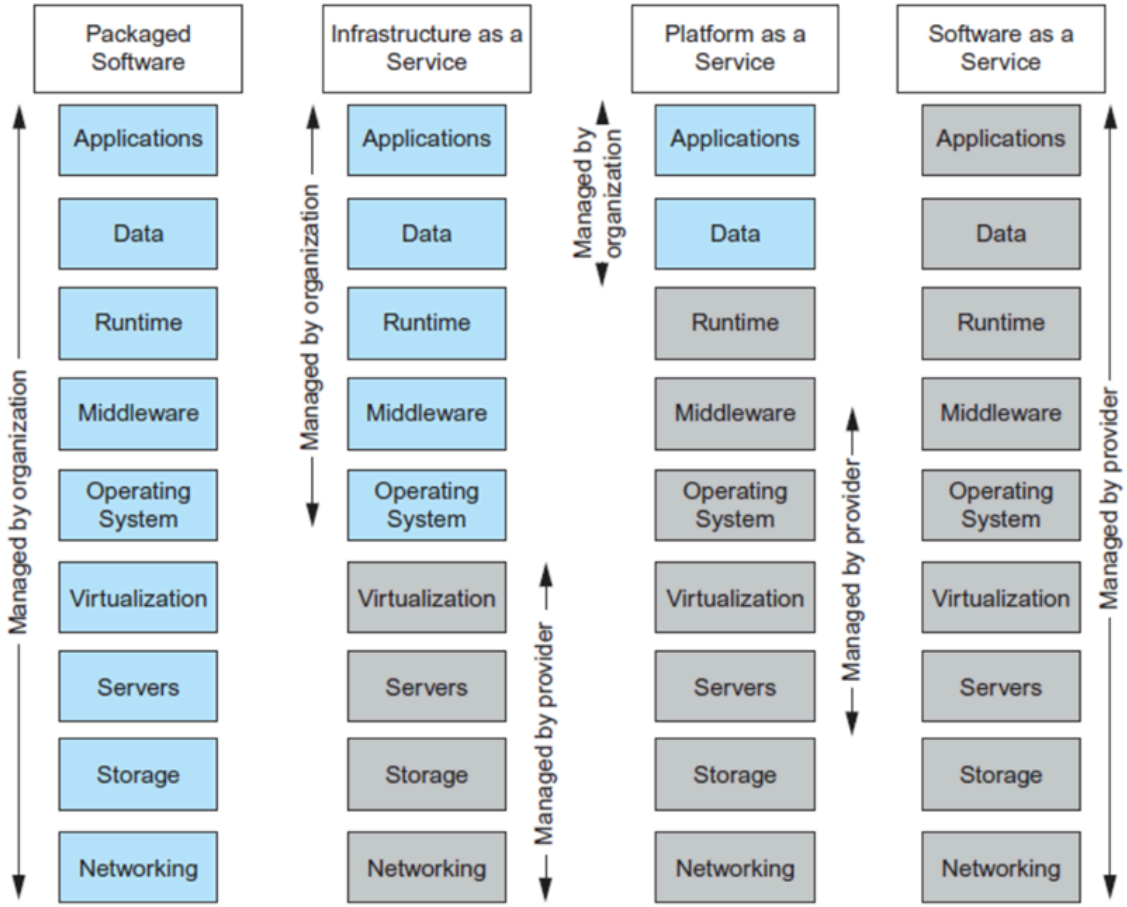
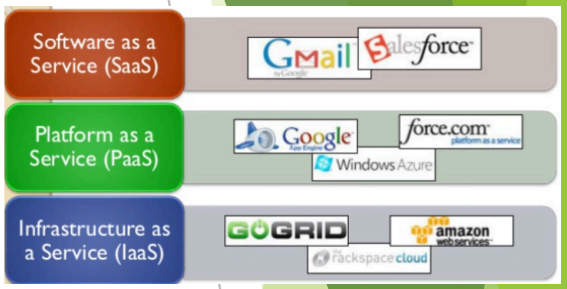
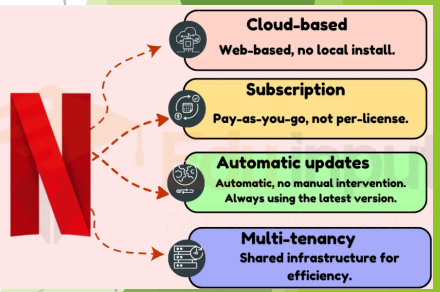


Figure 3.14 Differences between packaged software, IaaS, PaaS, and SaaS.





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https://cloudcomputinggate.com/examples-of-iaas-paas-and-saas/

Sources

- "Database Management Systems 1," T.ÖZSEVEN, Murathan Publications, 2010
- http://content.lms.sabis.sakarya.edu.tr/Uploads/71706/28378/veritabaniyonetimsistemleri21.pdf

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